

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Nerve net (1) Unmyelinated / branched/ bidirectional impulses / impulses travel in many directions (1)	2			2		
	(b)	(i)		Synaptic {knob / bulb} / axon terminal	1			1		
		(ii)		Vesicle: synaptic Accept secretory / excretory / neurosecretory and Role: transports {neurotransmitter / acetyl choline} Name and function for 1 mark	1			1		
		(iii)		{Acetylcholine / Neurotransmitter} {not broken down / remains in synaptic cleft} (1) Continues to stimulate {post synaptic membrane / adjacent cell}/ Continuous {depolarisation / action potentials} in {adjacent cell / neurone} / more {Acetylcholine / Neurotransmitter} binding to receptor on post-synaptic membrane (1)		2		2		
	(c)	(i)		X- depolarisation Y- repolarisation both for 1 mark	1			1		
		(ii)		(Na ⁺) Channel opens (1) {Na ⁺ / sodium ions} rush {into axon / into neurone / through membrane} (1) Accept flood/ rapidly diffuse in	2			2		